AI POWERED VR MOCK-INTERVIEW SIMULATION

A PROJECT REPORT

Submitted by

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BONAFIDE CERTIFICATE

Certified that this report titled **"AI-POWERED MOCK INTERVIEW SIMULATION"** is the bonafide work of **SWETHA N (241701059)** and **MUKESH VA (241701034)**, who carried out the work under my supervision. Certified further that, to the best of my knowledge, the work reported herein does not form part of any other thesis or dissertation on the basis of which a degree or award was conferred on an earlier occasion on these or any other candidates.

Certified further that the project contributes to **SDG 4 – Quality Education, SDG 9 – Industry, Innovation and Infrastructure, and SDG 8 – Decent Work and Economic Growth**, with a primary focus on SDG 4 – Quality Education. The project aims to support accessible, interactive, and personalised interview practice through a virtual reality-based mock interview platform that integrates conversational artificial intelligence and immersive simulation, helping students prepare for professional opportunities and develop career readiness.

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ABSTRACT

The **VR Simulated Mock Interview Platform** is designed to provide students with an immersive and personalised environment for interview preparation by integrating Virtual Reality (VR) and Artificial Intelligence (AI). The system extracts relevant information from candidates' PDF resumes to support personalised interview questions and conducts a sequential interview process involving three virtual interviewer roles: Human Resources (HR), behavioural interviewer, and hiring manager. Convai enables voice-based interaction with virtual characters, while Unity and OpenXR provide the virtual reality environment with desktop fallback support.

The system also analyses selected behavioural indicators, including head orientation, posture stability, and response latency, to generate practice-oriented performance feedback. At the end of the interview, a summary and PDF report are produced, and performance information can be synchronised with an institutional placement dashboard. The platform aims to support repeatable interview practice and help students identify areas for improvement. However, the evaluation methods are heuristic-based and require further validation, and formal user evaluation is necessary to establish the system's effectiveness.

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LIST OF ABBREVIATIONS

S.NO	ABBREVIATION	EXPANSION
1	AI	Artificial Intelligence
2	API	Application Programming Interface
3	CV	Curriculum Vitae
4	HMD	Head-Mounted Display
5	HR	Human Resources
6	HTTP	Hypertext Transfer Protocol
7	JSON	JavaScript Object Notation
8	LLM	Large Language Model
9	NLP	Natural Language Processing
10	OpenXR	Open Extended Reality
11	PDF	Portable Document Format
12	REST	Representational State Transfer
13	SDG	Sustainable Development Goals
14	UI	User Interface
15	VR	Virtual Reality
16	XR	Extended Reality

CHAPTER 1
INTRODUCTION

1.1 General

The development of Virtual Reality (VR) and Artificial Intelligence (AI) has created new opportunities for interactive learning and professional training. Interview preparation is an important part of students' career development, but traditional mock interviews may be limited by interviewer availability and the difficulty of providing consistent feedback. A realistic and repeatable interview practice environment can help students prepare for placement opportunities.

The proposed project, **"VR Interview Simulation: A Resume-Grounded, Sequential Multi-Persona Conversational AI Framework for Immersive Interview Practice,"** combines VR and conversational AI to provide personalised mock interviews based on the candidate's resume. The system includes three sequential virtual interviewers representing HR, behavioural assessment, and hiring-manager evaluation. It also analyses selected behavioural indicators and generates a performance feedback report, supporting structured interview practice for students.

1.2 Background

Virtual Reality (VR) and Artificial Intelligence (AI) are increasingly being used to create interactive and personalised learning experiences. In interview preparation, these technologies can provide a simulated environment where students practise answering questions and interacting with virtual interviewers. Conversational AI enables natural voice-based communication, while resume processing helps generate questions relevant to a candidate's skills, education, and projects.

The proposed system combines these technologies to develop a VR-based mock interview platform with three sequential interview stages: HR, behavioural assessment, and hiring-manager evaluation. It uses Unity for the virtual environment, Convai for conversational interactions, and a Python-based component for extracting information from resumes. The system also records selected behavioural indicators and generates feedback reports to support self-assessment and repeated practice. This provides a structured approach to

interview preparation, although the effectiveness of the evaluation methods requires further validation.

1.3 Problem Statement

Traditional interview preparation methods often lack personalised questions, realistic interview environments, and consistent performance feedback. Students may find it difficult to practise interviews with multiple interviewer roles and evaluate their communication and behavioural performance effectively.

Therefore, the project aims to develop a VR-based mock interview system that generates resume-grounded questions, enables sequential interactions with multiple AI interviewers, and provides automated feedback based on selected performance indicators. The system is intended to offer students a structured and interactive environment for repeated interview practice and self-assessment.

1.4 Motivation

The motivation behind this project is to provide students with an accessible and interactive platform for practising job interviews in a realistic virtual environment. Traditional mock interviews may not always provide opportunities to practise with different interviewer roles or receive consistent, personalised feedback. A VR-based simulation can help students become familiar with interview scenarios through repeated practice.

By integrating resume-based question generation, conversational AI, and automated performance feedback, the project aims to make interview preparation more personalised and structured. The system encourages students to identify areas for improvement and practise their interview skills before attending actual placement interviews.

1.5 Objectives

1. To develop an immersive virtual interview environment using Unity and OpenXR.
2. To extract relevant candidate information from PDF resumes for personalised question generation.
3. To integrate conversational AI for natural voice-based interaction with virtual interviewers.
4. To implement three sequential interview stages: HR, behavioural assessment, and hiring-manager evaluation.
5. To analyse selected behavioural indicators, including head orientation, posture stability, and response latency.
6. To generate automated performance feedback and downloadable PDF reports.
7. To integrate a placement dashboard for organised access to interview performance information.

1.6 Scope and Limitations

The project focuses on developing a VR-based mock interview platform that provides resume-based questions and interactive conversations with three sequential AI interviewers. It includes resume parsing, virtual interview simulation, behavioural indicator analysis, automated feedback generation, PDF report creation, and integration with an institutional placement dashboard. The system supports VR mode and desktop fallback, enabling students to practise interviews in a structured environment.

The system uses heuristic-based methods to evaluate interview performance, and the scoring approach has not been validated through formal user studies. Head orientation is used as an approximate indicator of attention and does not provide direct eye-tracking measurements. The effectiveness of the system in improving interview confidence or reducing anxiety has not yet been established. Advanced technical interview functionality is not implemented in the current version, and further testing and validation are required.

1.7 Existing System

Existing interview preparation systems mainly include traditional mock interviews, online interview platforms, and AI-based interview practice applications. Traditional mock interviews are conducted by faculty members, placement trainers, or peers, while online

platforms provide video-based interview practice. Some AI-based systems analyse candidate responses and provide feedback on communication skills and interview performance.

1.8 Limitations of Existing Approaches

The existing interview preparation systems have several limitations. Traditional mock interviews depend on the availability of interviewers and may not provide consistent feedback to every student. Online interview platforms often lack an immersive virtual environment and realistic interaction with multiple interviewer roles. Some AI-based platforms provide automated feedback but may offer limited personalisation based on the candidate's resume.

Additionally, existing approaches may not provide a continuous multi-stage interview experience or analyse behavioural indicators in an integrated manner. These limitations create a need for a system that combines resume-based question generation, conversational AI, immersive interview simulation, and structured performance feedback in a single platform.

1.9 Proposed System

The proposed VR Interview Simulation system consists of interconnected components for resume processing, virtual interview simulation, conversational AI, performance evaluation, and report generation. The candidate uploads a PDF resume, from which relevant information is extracted using a Python-based processing component. This information is used to personalise the interview questions according to the candidate's skills, education, and projects.

The system provides three sequential interview stages involving HR, behavioural, and hiring-manager virtual interviewers. Unity and OpenXR provide the virtual environment, while Convai enables voice-based conversations with virtual characters. Information from previous interview stages is passed to subsequent stages to maintain continuity. The system also records selected behavioural indicators, generates a performance summary and PDF report, and supports synchronisation with an institutional placement dashboard. A desktop fallback is available for users without a VR headset.

1.10 Advantages and Applications

The primary advantage of the proposed system is the integration of virtual reality and conversational AI to provide personalised and interactive interview practice. Resume-based question generation helps candidates practise questions related to their own profiles, while sequential virtual interviewers simulate different stages of an interview process. Automated feedback and PDF reports help users review their performance and identify areas for improvement.

The system can be applied in colleges, placement training centres, and career development programmes to support student interview preparation. The institutional placement dashboard provides an organised way to access interview performance information. The desktop fallback improves accessibility for users without VR equipment. However, the feedback scores are based on heuristic methods and require further validation before being used for formal candidate assessment.

CHAPTER 2
LITERATURE SURVEY

2.1 Introduction

Virtual Reality (VR) and Artificial Intelligence (AI) are emerging technologies used in interactive learning, simulation, and professional training. Existing approaches to interview preparation include traditional mock interviews, online interview platforms, conversational AI systems, and automated performance evaluation tools. Each approach offers different levels of interaction, personalisation, and feedback.

This literature survey examines research related to VR-based training, conversational AI, resume parsing, and behavioural performance analysis. These areas provide the theoretical foundation for developing the proposed VR Interview Simulation system, which combines resume-based question generation, sequential virtual interviewers, and automated feedback within a single platform.

2.2 Review of Related Research

Research on immersive training environments has explored the use of VR to simulate real-world situations and provide opportunities for repeated practice. In interview preparation, virtual environments can represent interview settings and allow users to interact with simulated interviewers. Such systems provide a controlled environment for practising communication and becoming familiar with interview procedures.

Conversational AI research focuses on enabling natural language interactions between users and computer systems. Voice-based AI characters can ask questions, receive responses, and maintain conversations, making them suitable for simulated interview scenarios. Integrating multiple AI interviewer roles can represent different interview stages and provide a more structured practice experience.

Resume parsing is another relevant research area, involving the extraction of information such as education, skills, projects, and experience from resumes. Structured candidate information can support personalised question generation. In addition, automated feedback systems can examine indicators such as response latency and head orientation to help users review their practice sessions. However, such indicators do not directly establish a candidate's confidence or communication ability without appropriate validation.

2.3 Existing Interview Preparation Technologies

Existing interview preparation technologies include online mock interview platforms, AI-based interview assistants, and virtual simulation systems. Online platforms allow candidates to practise answering interview questions, while AI-based tools can provide automated feedback on responses. Virtual simulation systems offer interactive environments that represent real-world situations.

However, these technologies differ in their levels of resume-based personalisation, immersive interaction, and multi-stage interview support. The proposed system integrates resume parsing, conversational AI, sequential virtual interviewers, and automated feedback to provide a structured interview practice experience. Its evaluation features are intended for practice and self-assessment and require further validation

2.4 Behavioural Analysis and Performance Feedback

Behavioural analysis is used in interview training systems to help candidates review their interaction patterns during practice sessions. Indicators such as head orientation, posture stability, and response latency can provide additional information about a candidate's behaviour during an interview. These indicators can be recorded and summarised to support self-assessment.

Automated feedback systems organise performance information into understandable summaries and reports. Such feedback helps candidates identify areas that may require further practice. However, behavioural indicators are approximate measures and cannot independently determine a candidate's confidence, attention, or communication ability.

The proposed VR Interview Simulation system records selected behavioural indicators during interview sessions and combines them with heuristic-based evaluation to generate feedback. The results are presented through an on-screen summary and a PDF report. These features are intended to support interview practice rather than provide validated assessments for recruitment decisions.

t and require further validation

2.5 Resume-Based Question Generation and Conversational AI

Resume-based question generation involves extracting information such as educational qualifications, technical skills, projects, and experience from a candidate's resume. This

information can be organised into a structured profile and used to generate relevant interview questions. Personalised questions help candidates practise discussing their own qualifications and project experiences.

Conversational AI enables voice-based interaction between candidates and virtual interviewers. AI-powered characters can ask questions, receive responses, and maintain a conversation throughout an interview session. Passing information between sequential interview stages helps maintain continuity across different interviewer roles.

The proposed system combines local PDF resume processing with Convai-based conversational interactions. The extracted candidate profile supports personalised interview preparation, while three sequential interviewer roles represent HR, behavioural assessment, and hiring-manager evaluation. The system is designed to provide a structured interview experience, although the quality and relevance of generated questions may depend on the accuracy of resume extraction and conversational AI responses.

Table 2.1: Comparative Analysis of Existing Interview Preparation Systems

System	Primary Function	Key Features
Traditional Mock Interviews	Face-to-face interview practice	Direct interaction, verbal feedback, interviewer guidance
Online interview platforms	Virtual interview practice	Virtual interview practicen banks, remote access
AI-Based Interview Tools	Automated interview preparation	AI-generated questions, response analysis, automated feedback
AI-generated questions, response analysis, automated feedback	Immersive simulation	Virtual environments, interactive scenarios, realistic practice
Proposed VR Interview Simulation	Personalised immersive interview practice	Resume-based questions, three sequential AI interviewers, behavioural indicators, PDF feedback reports

CHAPTER 3
SYSTEM DESIGN

3.1 General

System design establishes the architecture, components and communication mechanisms required for integrating resume processing, virtual interview simulation and conversational AI. The proposed VR Interview Simulation consists of interconnected components, including the Unity-based virtual environment, a Python-based resume processing module, Convai-powered virtual interviewers and a placement dashboard. The resume processing module extracts relevant candidate information, while the virtual environment provides an interactive interview setting. Conversational AI enables voice-based interaction with three sequential interviewer roles, and the feedback module generates performance summaries and reports.

3.2 Requirement Analysis

The primary requirement of the system is to provide personalised interview practice by extracting information from candidate resumes, conducting sequential AI-based interviews and generating performance feedback. The system must support PDF resume upload, candidate profile extraction, voice-based interaction, and the transfer of relevant information between interview stages. It must also record selected behavioural indicators and generate a PDF report. These requirements are categorised into functional requirements, which define the operations performed by the system, and non-functional requirements, which address usability, reliability, responsiveness, maintainability and accessibility.

3.3 Functional and Non-Functional Requirements

The requirements of the VR Interview Simulation system are divided into functional and non-functional categories to ensure that the platform provides the necessary capabilities while maintaining practical operating characteristics.

3.3.1 Functional Requirements

The system must allow users to upload PDF resumes and extract relevant information such as education, skills, projects and experience. The Unity application must provide an interview environment with three sequential stages: HR, behavioural assessment and hiring-manager evaluation. Convai must support voice-based conversations with virtual interviewers, while relevant information from previous stages is passed to subsequent stages. The system must record selected behavioural indicators, generate a performance summary, create a downloadable PDF report and support synchronisation with the institutional placement dashboard.

3.3.2 Non-Functional Requirements

The non-functional requirements focus on usability, reliability, responsiveness, accessibility and maintainability. The system should provide a clear interface for resume upload, interview initiation and feedback review. Interview interactions should remain responsive, and the system should handle resume-processing and communication errors appropriately. Modular separation of the Unity application, Python processing component, conversational AI integration and dashboard services should support testing and future improvements. The desktop fallback provides an alternative for users without VR equipment. The feedback mechanism uses heuristic-based evaluation and is intended for practice rather than validated recruitment assessment.

3.4 Overall System Architecture

The VR Interview Simulation follows a connected software architecture consisting of resume processing, virtual interview simulation, conversational AI, behavioural data collection, feedback generation and dashboard integration. When a candidate uploads a PDF resume, the Python-based processing component extracts relevant information and prepares a structured candidate profile. The profile is then used to personalise the interview experience. Unity provides the virtual environment, while Convai enables interactions with virtual interviewers through voice-based conversations.

The interview proceeds through three sequential stages involving HR, behavioural and hiring-manager roles. Information from earlier stages is passed to subsequent interviewers to maintain continuity. During the session, selected indicators such as head orientation, posture stability and response latency are recorded. After the interview, the system

generates a performance summary and PDF report and supports transferring performance information to the institutional placement dashboard. This architecture separates resume processing, interview interaction, performance evaluation and reporting into distinct components, supporting organised development and future enhancements.

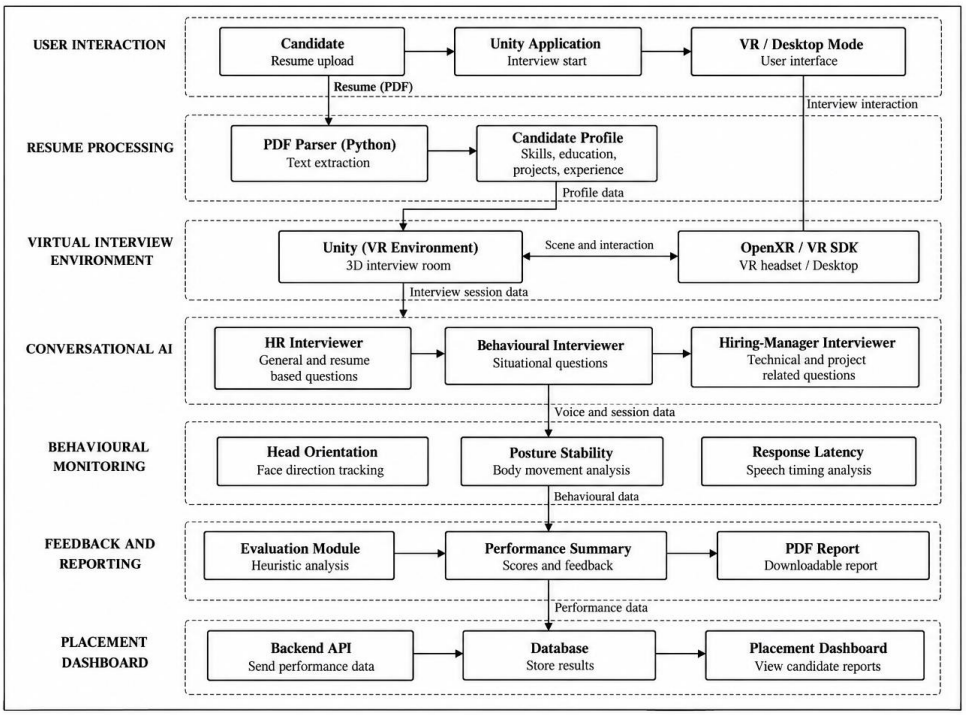


Figure 3.1 Overall system architecture of the Proposed VR Interview Simulation System

Figure 3.1: Overall system architecture of VR Interview Platform

3.5 Software Integration

Software integration connects the resume-processing component, Unity-based virtual environment, conversational AI, behavioural monitoring, feedback generation and placement dashboard. The Python component extracts relevant information from uploaded PDF resumes and prepares a structured candidate profile. This information is used to personalise the interview experience within Unity.

Convai enables voice-based interaction with three sequential virtual interviewers representing HR, behavioural assessment and hiring-manager evaluation. Information from previous interview stages is passed to subsequent stages to maintain continuity. Selected behavioural indicators are collected during the session and processed by the evaluation component to generate a performance summary and PDF report. The system also supports synchronisation with the institutional placement dashboard through a backend API

3.6 System Flow Diagram

The system workflow begins when the candidate uploads a PDF resume and starts an interview session. The Python-based resume-processing component extracts relevant information and prepares a structured candidate profile. Unity then launches the virtual interview environment, where the candidate interacts with the HR, behavioural and hiring-manager interviewers in sequence through Convai.

During the interview, the system records selected behavioural indicators, including head orientation, posture stability and response latency. After the session, the evaluation component generates a performance summary and downloadable PDF report. The resulting information can also be synchronised with the placement dashboard for organised review.

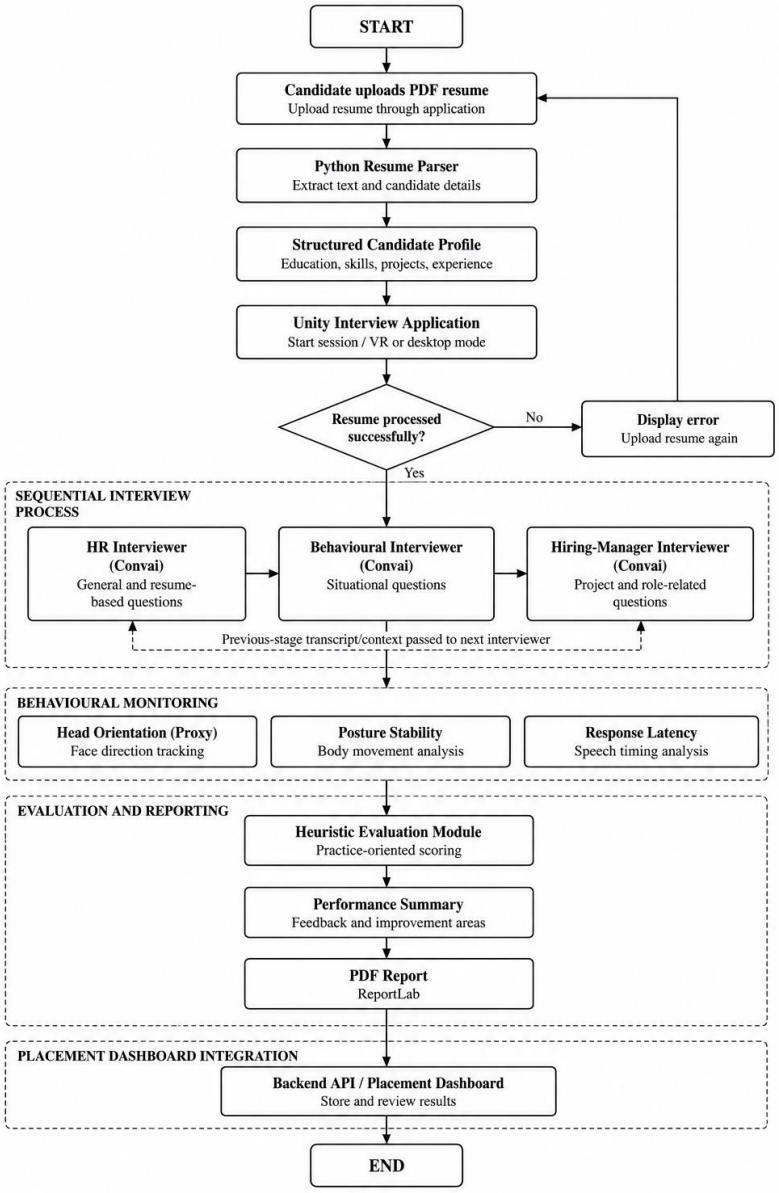


Figure 3.2 System flow diagram of the proposed VR Interview Simulation System

Figure 3.2: Overall system flow diagram

3.7 Use Case Diagram

The use case diagram represents the interactions between the candidate and the VR Interview Simulation system. The candidate can upload a resume, start an interview, interact with virtual interviewers, complete the sequential interview stages and view performance feedback. The system processes the resume, maintains continuity between interviewer stages, records selected behavioural indicators and generates a report.

The placement dashboard provides an additional interface for accessing interview performance information. The use case diagram illustrates the relationship between the candidate, the interview simulation and the supporting software components.

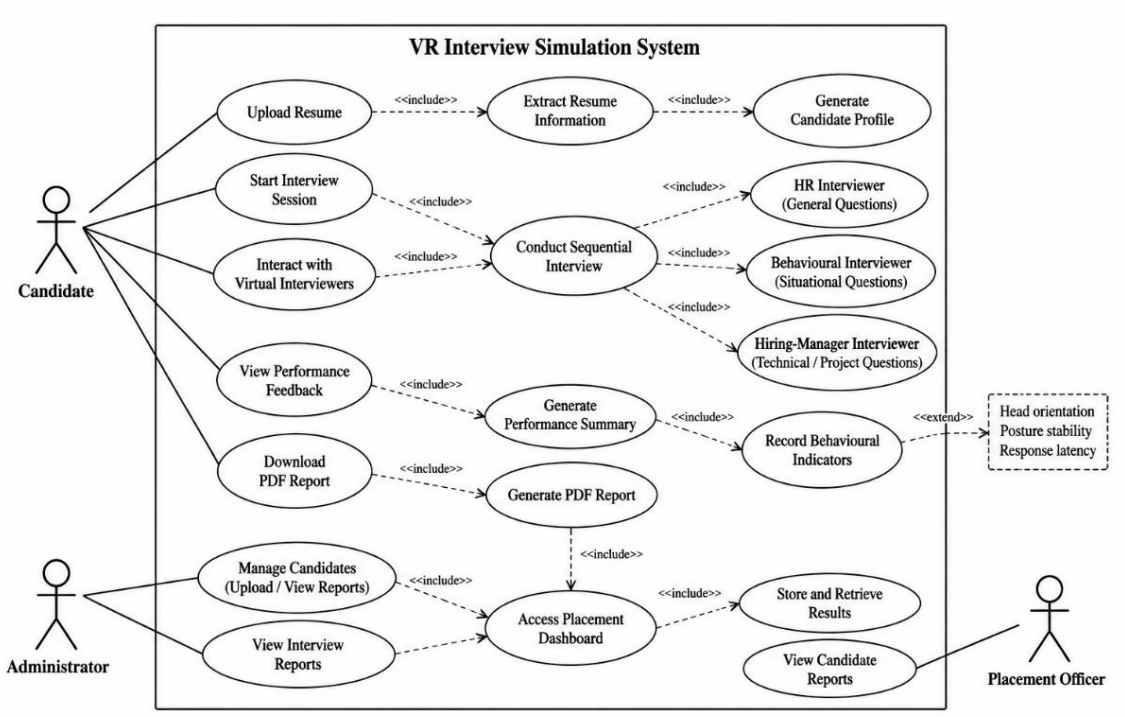


Figure 3.3 Use case diagram of the proposed VR Interview Simulation System

Figure 3.3: Use case diagram of the Proposed System

3.8 Activity and Sequence Diagrams

Activity and sequence diagrams describe the workflow and communication between the major components of the proposed VR Interview Simulation system. The activity diagram represents the steps involved in resume processing, interview execution and feedback generation, while the sequence diagram illustrates how information is exchanged between the candidate, Unity application, Python resume parser, Convai service and backend components.

3.8.1 Activity Diagram

An activity diagram describes the sequence of operations and decisions involved in a particular workflow. In the proposed system, the activity begins when the candidate uploads a PDF resume. The Python-based resume parser extracts relevant information and prepares a structured candidate profile. The candidate then starts the interview session in the Unity application, which provides the virtual interview environment.

The interview proceeds through three sequential stages: HR, behavioural assessment and hiring-manager evaluation. Information from previous stages is passed to subsequent interviewers to maintain continuity. During the session, selected behavioural indicators,

including head orientation, posture stability and response latency, are recorded. After the interview is completed, the evaluation component generates a performance summary and PDF report. The results can also be synchronised with the placement dashboard. This diagram illustrates the complete workflow from resume upload to final feedback generation.

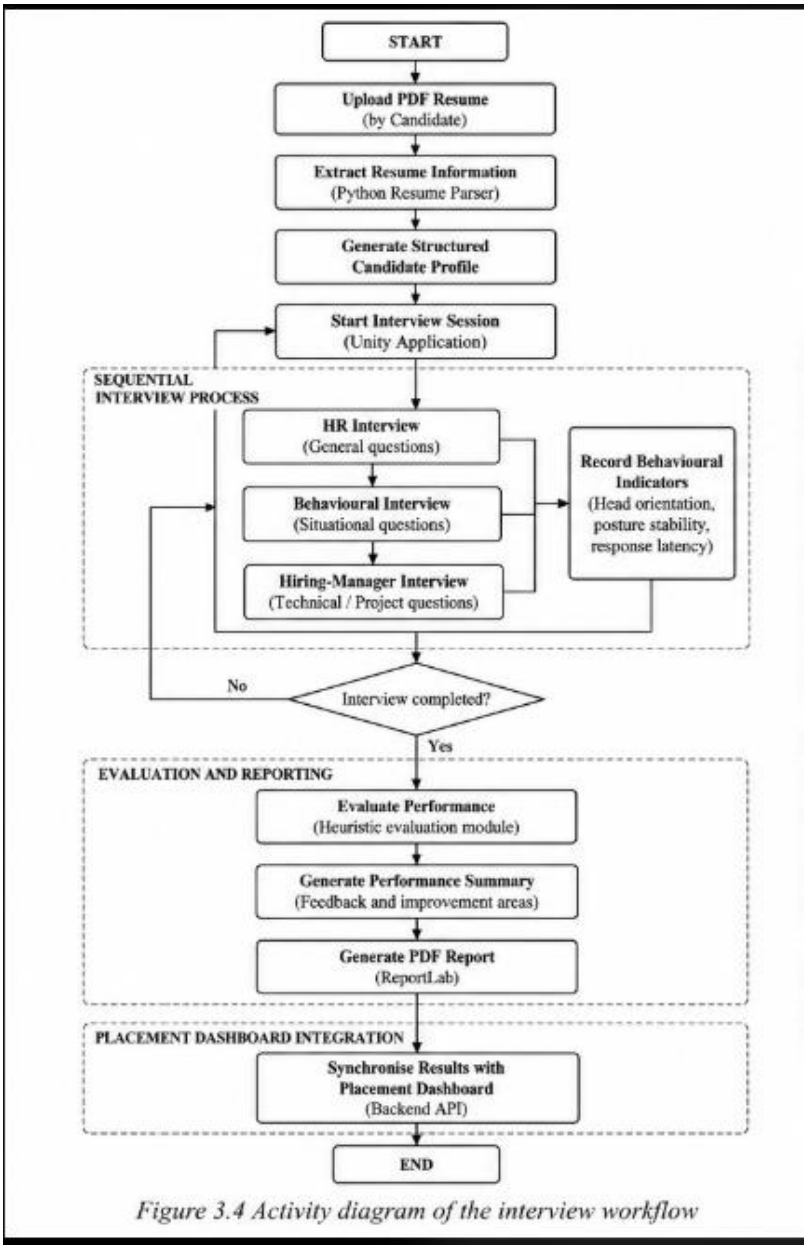


Figure 3.4: Activity Diagram of the Interview Workflow

3.8.2 Sequence Diagram

A sequence diagram illustrates the exchange of information between system components in the order in which communication occurs. In the proposed VR Interview Simulation system, the main participating components are the candidate, Unity application, Python

resume parser, Convai service, evaluation module and backend API. The process begins when the candidate uploads a PDF resume. The Python component extracts relevant information and prepares a structured candidate profile, which is used by the application to personalise the interview session.

During the interview, Unity communicates with Convai to enable voice-based interaction with the virtual interviewers. The interview proceeds through the HR, behavioural and hiring-manager stages. Information from previous stages is passed to subsequent interviewers to maintain continuity. Selected behavioural indicators are recorded during the session, and the available interview data is processed by the evaluation module to generate a performance summary and PDF report. The backend API supports transferring the results to the placement dashboard.

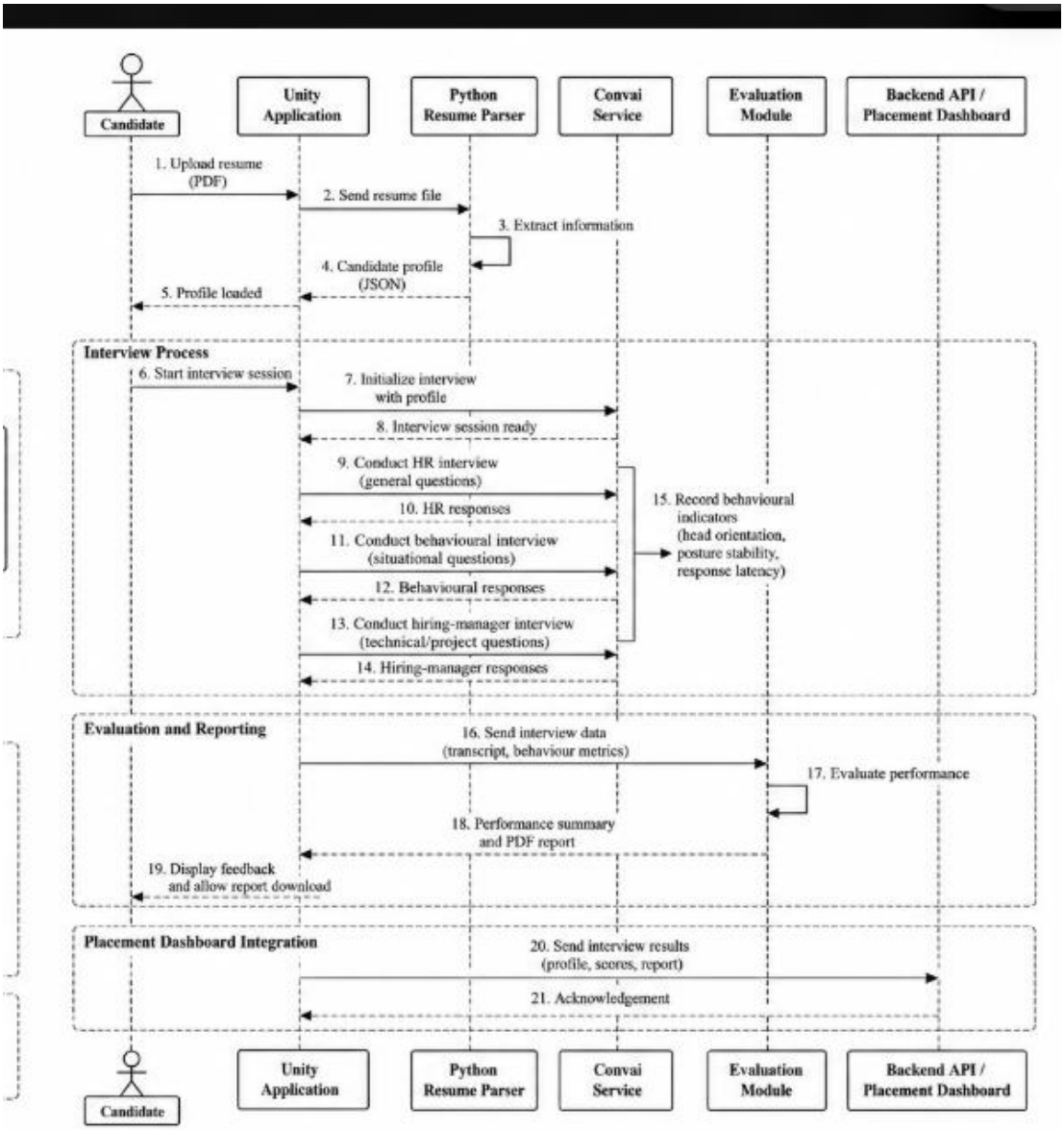


Figure 3.5 Sequence diagram of the proposed VR Interview Simulation System

Figure 3.5: Sequence Diagram of the Interview Workflow

3.9 Module Design

The VR Interview Simulation system is divided into several software modules, each responsible for a specific function. These modules work together to provide a personalised and interactive interview experience. The major modules include resume processing, virtual interview simulation, conversational AI, behavioural monitoring, performance evaluation, report generation and placement dashboard integration.

1. **Resume Processing Module**

This module accepts PDF resumes and extracts relevant information, including educational qualifications, technical skills, projects and experience. The extracted information is organised into a structured candidate profile for personalised interview preparation.

2. **Virtual Interview Module**

Developed using Unity and OpenXR, this module provides the virtual interview environment. It supports VR mode and desktop fallback, allowing candidates to participate in simulated interview sessions.

3. **Conversational AI Module**

This module integrates Convai to enable voice-based interactions with virtual interviewers. It supports three sequential interview stages: HR, behavioural assessment and hiring-manager evaluation. Information from previous stages is passed to subsequent stages to maintain continuity.

4. **Behavioural Monitoring Module**

This module records selected behavioural indicators during the interview, including head orientation, posture stability and response latency. These indicators provide additional information for reviewing the practice session but are not validated measures of interview performance.

5. **Performance Evaluation and Reporting Module**

This module processes available interview information using heuristic-based evaluation methods. It generates a performance summary highlighting feedback and potential areas for improvement. A downloadable PDF report is created using ReportLab.

6. **Backend and Placement Dashboard Module**

This module supports communication between the interview application and the backend services. Using the backend API, interview results can be synchronised with the institutional placement dashboard for organised storage and review.

CHAPTER 4
SOFTWARE IMPLEMENTATION

4.1 Software Architecture and Technology Stack

The software architecture of the proposed VR Interview Simulation system integrates resume processing, virtual reality, conversational artificial intelligence, behavioural monitoring and performance reporting. The system consists of several interconnected software components that work together to provide a personalised interview practice experience. Unity 6 provides the virtual interview environment, while OpenXR supports VR functionality and desktop fallback. A Python-based resume-processing component extracts relevant information from uploaded PDF resumes and prepares a structured candidate profile.

Convai provides conversational interactions with virtual interviewers, while the evaluation component processes selected interview information to generate performance feedback. ReportLab is used to generate downloadable PDF reports. The backend, developed using Node.js and Express, supports communication with the institutional placement dashboard, which uses Next.js for its interface and PostgreSQL for data storage through Prisma. The modular architecture separates resume processing, interview execution, evaluation and reporting to support maintenance and future development.

4.2 Unity Application and Virtual Interview Environment

The Unity application provides the main interface for candidates to participate in simulated interview sessions. Developed using Unity 6 and OpenXR, the application supports an immersive VR environment with a desktop fallback. The candidate can initiate an interview session and interact with virtual interviewers through voice-based communication.

The interview is organised into three sequential stages: HR, behavioural assessment and hiring-manager evaluation. Each stage represents a different interview role and focuses on relevant questions. Information from earlier stages is passed to subsequent interviewers to maintain continuity throughout the session. This structure provides a consistent workflow for practising different aspects of an interview.

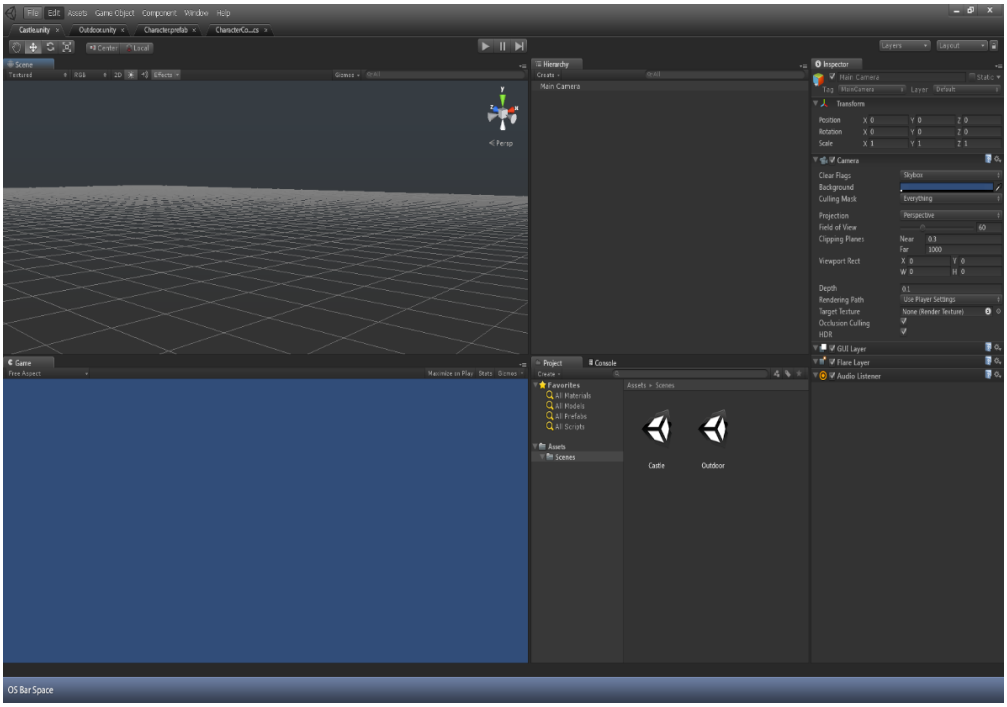


Figure 4.1: Unity Environment

4.3 Resume Processing Module

The resume-processing module accepts PDF resumes and extracts relevant candidate information using a Python-based processing component. The extracted information may include educational qualifications, technical skills, projects and experience. The module uses PDF text extraction and heuristic-based parsing to organise the available information into a structured candidate profile.

The generated profile is used to support personalised interview questions related to the candidate's background. Processing the resume before the interview helps connect the interview session with the candidate's documented experience. The accuracy of the extracted information depends on the PDF's text structure and the effectiveness of the parsing rules.

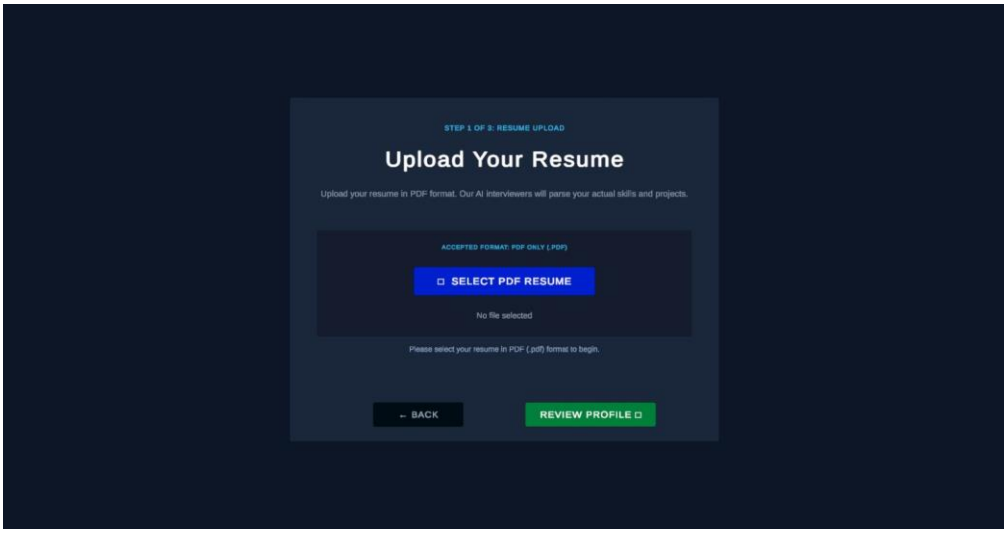


Figure 4.2: Resume Selection Interface

4.4 Conversational AI and Sequential Interviewing

The conversational AI module integrates Convai to enable voice-based interactions between candidates and virtual interviewers. The system includes three interviewer roles: HR interviewer, behavioural interviewer and hiring-manager interviewer. Each role represents a separate stage of the interview process.

The HR stage focuses on general and resume-related questions, while the behavioural stage focuses on situational questions. The hiring-manager stage focuses on projects and role-related topics. Information from previous interview stages is passed forward to maintain conversational continuity. The implemented interview stages provide a structured simulation of a multi-interviewer interview process.

4.5 Behavioural Monitoring

The behavioural monitoring module records selected indicators during the interview session. These include head orientation, posture stability and response latency. Head orientation is used as an approximate indicator of whether the candidate's head is directed towards the interviewer, while posture stability and response latency provide additional information about interaction patterns.

The collected indicators are used as inputs to the system's heuristic-based feedback process. However, head orientation does not directly measure eye movement, and the recorded indicators cannot independently establish a candidate's confidence or communication

ability. The module is intended to support interview practice and self-review rather than provide a validated psychological or recruitment assessment.



Figure 4.3: Unity Interview Environment

4.6 Performance Evaluation and Report Generation

The performance evaluation module processes available interview information and applies heuristic-based methods across ten competencies to produce a structured performance summary. The feedback is intended to help candidates identify areas that may require further practice.

The reporting component uses ReportLab to generate a downloadable PDF report containing the evaluation results. This allows candidates to review their interview performance after completing the session. Since the scoring method has not been validated through formal user studies, the generated scores should be interpreted as practice-oriented feedback rather than definitive measures of interview ability.

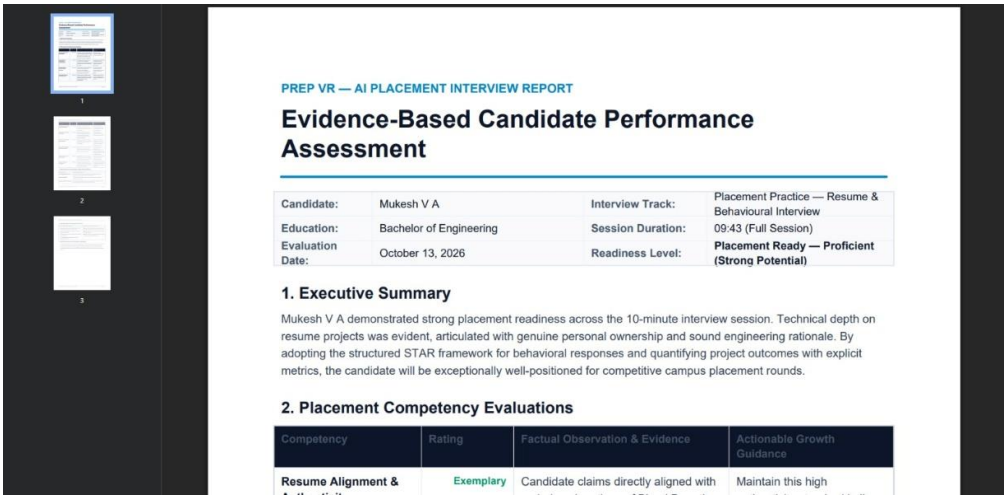


Figure 4.4: Feedback Obtained

4.7 Placement Dashboard Integration

The placement dashboard integration module connects the interview system with the institutional reporting interface through a backend API. The backend uses Node.js and Express, with Prisma and PostgreSQL supporting data management. The dashboard interface is developed using Next.js and provides an organised way to access interview-related information.

After the interview, the system supports synchronisation of available results with the placement dashboard. This integration connects the interview application with institutional review workflows and provides a basis for managing candidate performance information. The actual information available on the dashboard depends on the data transferred.

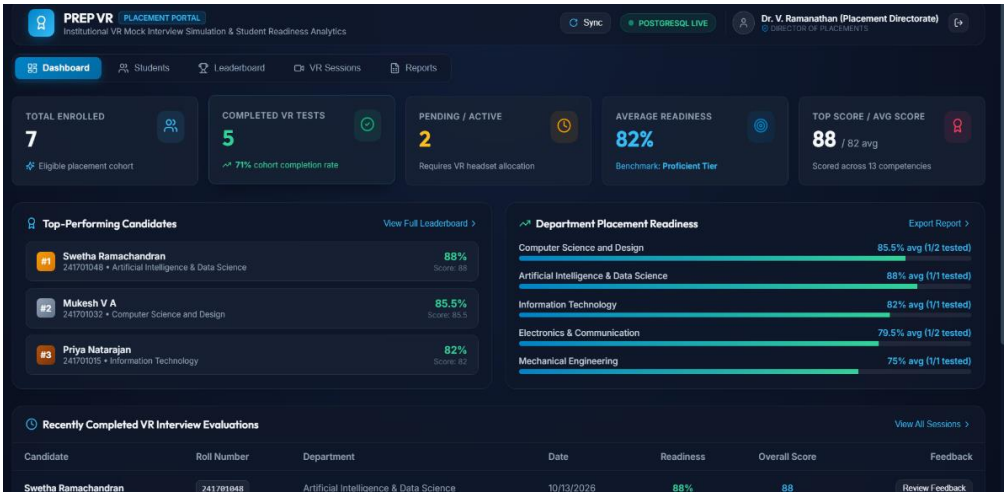


Figure 4.5: Placement Dashboard

CHAPTER 5
VR INTEGRATION

5.1 Virtual Reality Environment Setup

The virtual reality environment is developed using Unity 6 to provide an immersive and interactive interview experience. The environment simulates a professional interview setting where candidates can practise answering questions posed by virtual interviewers. The virtual environment is designed to help candidates become familiar with interview situations and improve their confidence through repeated practice.

The interview environment contains virtual interviewer characters and the necessary interaction elements to conduct the interview. The system supports a VR mode using OpenXR and also provides a desktop fallback mode. This allows the application to be accessed in different runtime configurations, depending on the available equipment.

5.2 Unity and OpenXR Integration

Unity 6 serves as the primary development platform for creating the virtual interview environment and managing the interview workflow. OpenXR is integrated to support virtual reality functionality and communication with compatible VR devices. The integration enables the application to operate in a VR environment while maintaining a desktop fallback option.

The Unity application manages the interview scene, virtual characters, user interactions, and transitions between interview stages. The XR Interaction Toolkit supports VR-related interaction functionality. This integration forms the foundation for connecting the virtual environment with the conversational AI system and other backend components.

5.3 VR Interaction and Interview Navigation

The system allows candidates to participate in a simulated interview through interactions with virtual interviewer characters. The interview is organised into three sequential stages: Human Resources (HR), behavioural interview, and hiring-manager interview. Each stage is designed to represent a different aspect of a real interview process.

Each interview stage runs for approximately 200 seconds, giving a total planned interview duration of 600 seconds. After one stage is completed, the system transitions to the next interviewer. The previous stage's transcript is forwarded to the subsequent stage to maintain conversational continuity. This sequential structure provides candidates with an organised interview experience within the virtual environment.

5.4 Integration of Conversational AI with the VR Environment

The VR environment is integrated with Convai to enable voice-based conversations with virtual interviewer characters. Convai provides the conversational interaction functionality, allowing the virtual interviewers to ask questions and respond during the interview session. The integration connects the immersive environment with cloud-hosted conversational AI services.

The system uses three virtual interviewer stages corresponding to HR, behavioural, and hiring-manager interviews. The conversation progresses sequentially, and transcripts from earlier stages are passed to later stages to provide context. Candidate resume information is processed separately by the local Python-based resume-processing component and structured into a candidate profile for use in the interview workflow.

After the interview, the system evaluates selected performance indicators using heuristic-based methods and generates a PDF report. The results can also be synchronised with the institutional placement dashboard through a REST API. The overall integration combines virtual reality, conversational AI, resume processing, performance evaluation, and reporting into a single interview-practice framework.

Note: The behavioural scores are engineering heuristics and have not been validated through user studies. Head orientation is used as a proxy for head alignment, not as direct eye tracking.

CHAPTER 6
METHODOLOGY AND ALGORITHMS

6.1 Methodology

The proposed VR Interview Simulation system follows a systematic methodology that combines resume processing, virtual reality, conversational artificial intelligence, behavioural monitoring, performance evaluation, and report generation. The methodology is divided into the following stages.

6.1.1 Resume Upload and Processing

The candidate uploads a resume in PDF format. A local Python component extracts the text using the pypdf library. Regular expressions and keyword-based heuristics are applied to identify relevant candidate information, such as skills, educational qualifications, and project experience. The extracted information is organised into a structured candidate profile for use in the interview workflow.

6.1.2 Virtual Interview Environment Initialization

The virtual interview environment is developed using Unity 6 with OpenXR support. The environment is initialised with virtual interviewer characters and the necessary interaction components. A desktop fallback mode is also provided to support application execution without VR mode.

6.1.3 Sequential AI-Based Interview

The interview is conducted in three sequential stages:

1. HR Interview: Focuses on the candidate's introduction, background, and career-related questions.
2. Behavioural Interview: Focuses on previous experiences, problem-solving approaches, and behavioural responses.
3. Hiring-Manager Interview: Focuses on skills, project experience, and role-related questions.

Convai provides conversational interaction with the virtual interviewer characters. Each stage runs for approximately 200 seconds, resulting in a total planned interview duration

of 600 seconds. Transcripts from preceding stages are forwarded to subsequent stages to maintain conversational continuity.

6.1.4 Behavioural Monitoring

During the interview, the system samples the VR headset's head pose at a frequency of 1 Hz. A heuristic 35-degree cone is used to estimate head alignment with the interviewer. Posture stability and response latency are also considered as behavioural indicators.

These measurements provide approximate indicators of interview behaviour. Head orientation is used as a proxy for head alignment and does not represent direct eye tracking.

6.1.5 Performance Evaluation

After the interview, the system evaluates the available interview information using a heuristic-based approach across ten competencies. The evaluation produces a preliminary performance summary intended to help candidates understand their interview performance and identify areas for improvement.

The evaluation scores are engineering heuristics and have not been validated through formal user studies. Therefore, they should be interpreted as practice feedback rather than definitive measurements of interview ability.

6.1.6 Report Generation and Dashboard Integration

The evaluation results are presented through an on-screen summary. A PDF report is generated using the ReportLab library. The relevant results can also be synchronised with the institutional placement dashboard through a REST API, allowing interview performance information to be accessed through the dashboard.

6.2 Algorithms

6.2.1 Resume Parsing Algorithm

Purpose: To extract relevant candidate information from a PDF resume.

Algorithm:

1. Start.
2. Accept the PDF resume uploaded by the candidate.
3. Extract the text using the pypdf library.
4. Apply regular expressions and keyword-based heuristics to identify relevant information.
5. Organise the extracted information into a structured candidate profile.
6. Make the profile available to the interview workflow.
7. Stop.

Output: Structured candidate profile containing the information extracted from the resume.

6.2.2 Sequential Interview Algorithm

Purpose: To conduct a multi-stage interview using virtual AI interviewers.

Algorithm:

1. Start the interview session.
2. Initialise the HR interviewer.
3. Conduct the HR interview for approximately 200 seconds.
4. Collect and store the HR interview transcript.
5. Forward the available transcript context to the behavioural interviewer.
6. Conduct the behavioural interview for approximately 200 seconds.
7. Forward the available interview context to the hiring-manager interviewer.
8. Conduct the hiring-manager interview for approximately 200 seconds.
9. Collect the transcripts from all three stages.
10. Pass the available interview results to the evaluation module.
11. Stop.

Output: Interview transcripts from the three sequential interview stages.

6.2.3 Behavioural Monitoring Algorithm

Purpose: To derive approximate behavioural indicators from VR headset data.

Algorithm:

1. Start behavioural monitoring during the interview.
2. Sample the headset's head pose at 1 Hz.
3. Obtain the available head-orientation information.
4. Apply the heuristic 35-degree cone to estimate head alignment with the interviewer.
5. Analyse posture stability using the available pose information.
6. Record response latency where applicable.
7. Provide the resulting indicators to the evaluation module.
8. Stop monitoring when the interview ends.

Output: Approximate head-alignment, posture-stability, and response-latency indicators.

6.2.4 Performance Evaluation Algorithm

Purpose: To generate a preliminary interview performance summary.

Algorithm:

1. Start.
2. Retrieve the available interview transcripts.
3. Retrieve the behavioural indicators collected during the interview.
4. Evaluate the available information using the implemented heuristic rules.
5. Produce scores or indicators across ten competencies.
6. Generate the performance summary according to the implemented evaluation logic.
7. Forward the results to the reporting module.
8. Stop.

Output: Preliminary competency-based performance summary.

6.2.5 Report Generation Algorithm

Purpose: To generate a PDF report containing the interview evaluation results.

Algorithm:

1. Start.
2. Receive the evaluation results.
3. Organise the available competency scores and performance summary.
4. Generate the PDF report using ReportLab.
5. Make the report available to the candidate.
6. Synchronise the relevant results with the placement dashboard through the REST API.
7. Stop.

Output: PDF interview evaluation report and dashboard-synchronised results.

6.3 Summary

The methodology integrates resume processing, virtual reality, conversational AI, behavioural monitoring, heuristic-based evaluation, and report generation into a unified interview-practice workflow. The algorithms describe the main processing stages of the system, from resume upload to the generation of interview feedback. The proposed framework provides a structured environment for repeated interview practice while recognising the limitations of heuristic-based behavioural and performance measurements.

CHAPTER 7
TESTING AND RESULTS

7.1 Introduction

Testing is an important phase in the development of the VR Interview Simulation system. It is performed to verify the functionality of the major modules and ensure that the components work together as expected. The testing process focuses on resume processing, virtual interview execution, conversational AI integration, behavioural monitoring, performance evaluation, PDF report generation, and placement dashboard integration.

7.2 Testing Methodology

The system is evaluated through module-level checks and integration testing. Each major component is considered separately to verify its intended functionality. Integration testing focuses on the interaction between the Unity application, Python-based resume processing component, Convai conversational AI service, reporting module, and placement dashboard backend.

The testing methodology includes the following areas:

1. Resume processing and candidate profile creation.
2. Initialisation of the virtual interview environment.
3. Execution of the three sequential interview stages.
4. Collection of headset pose data and behavioural indicators.
5. Generation of competency-based evaluation results.
6. Creation of PDF reports.
7. Communication with the placement dashboard through the REST API.

7.3 Functional Testing

Functional testing verifies whether each module performs its intended operation.

7.3.1 Resume Processing Testing

The resume processing module is designed to accept PDF resumes, extract textual content, and identify relevant candidate information using regular expressions and keyword-based

heuristics. The extracted information is organised into a structured candidate profile for use in the interview workflow.

7.3.2 Virtual Reality Testing

The virtual interview environment is developed using Unity 6 with OpenXR support. Testing focuses on initialising the interview scene, displaying virtual interviewer characters, and supporting the interview workflow. A desktop fallback mode is also available.

7.3.3 Conversational AI Testing

The conversational AI integration uses Convai to support interactions with virtual interviewer characters. The interview workflow consists of three stages: HR, behavioural, and hiring-manager interviews. Each stage is designed to run for approximately 200 seconds, with context from preceding interview transcripts forwarded to subsequent stages.

7.3.4 Behavioural Monitoring Testing

The behavioural monitoring component samples headset pose information at 1 Hz. A heuristic 35-degree cone is used to estimate head alignment with the interviewer. Posture stability and response latency are also considered as behavioural indicators. These measurements provide approximate indicators and do not constitute direct eye tracking.

7.3.5 Report Generation Testing

The reporting module uses the available evaluation results to generate a PDF report through the ReportLab library. The generated report is intended to present the candidate's interview performance summary and competency-based feedback.

7.3.6 Dashboard Integration Testing

The system supports synchronisation of relevant interview results with the institutional placement dashboard through a REST API. Integration testing should verify that the expected data is transmitted and handled correctly by the backend.

7.4 Integration Testing

Integration testing examines the interaction between the different modules of the proposed system. The major integration paths include resume processing with the interview

workflow, Unity with Convai, interview transcript forwarding between stages, evaluation with report generation, and report-related data synchronisation with the placement dashboard.

The purpose of integration testing is to identify communication errors, missing data, and workflow interruptions that may occur when individual modules operate together.

7.5 Results and Discussion

The proposed system brings together resume processing, a virtual interview environment, conversational AI, behavioural monitoring, performance evaluation, and report generation in a unified interview-practice workflow.

The design supports three sequential interview stages and provides a mechanism for carrying interview context between stages. Headset pose information is used to derive approximate behavioural indicators, while the evaluation module produces a preliminary competency-based performance summary. The reporting component generates a PDF report, and the backend supports synchronisation with the placement dashboard.

However, the evaluation scores are based on engineering heuristics and have not been validated through formal user studies. Therefore, the system's ability to improve interview confidence, reduce anxiety, or predict actual interview performance has not been established.

7.6 Limitations of Testing

The current evaluation focuses on the technical functionality and integration of the system. Formal user studies have not been conducted to establish the effectiveness of the proposed interview-practice framework. The behavioural indicators are approximate, and head orientation cannot be treated as direct eye tracking. The heuristic scoring mechanism also requires further validation before it can be used for reliable assessment.

The technical coding interview track is not implemented in the current version. Further testing is required to assess usability, reliability, scoring consistency, and the effectiveness of the system in realistic interview-practice scenarios.

7.7 Summary

This chapter described the testing approach and the intended functionality of the major modules in the VR Interview Simulation system. The testing areas include resume processing, virtual reality integration, conversational AI, behavioural monitoring, performance evaluation, PDF report generation, and dashboard integration. Formal user evaluation and validation of the heuristic scoring mechanism remain important areas for future work.

CHAPTER 8
CONCLUSION AND FUTURE SCOPE

8.1 Conclusion

The VR Interview Simulation system demonstrates the development of an integrated virtual reality and conversational artificial intelligence framework for immersive interview practice. The system combines a Unity 6 virtual environment, OpenXR support, Convai-based virtual interviewer characters, and a Python-based resume-processing component to provide a structured interview workflow. The resume-processing module extracts relevant information from PDF resumes and organises it into a structured candidate profile. The interview is conducted through three sequential stages: HR, behavioural, and hiring-manager interviews, with transcripts from earlier stages forwarded to subsequent stages to maintain conversational continuity. The system also incorporates headset pose monitoring to derive approximate behavioural indicators, heuristic-based evaluation across ten competencies, PDF report generation using ReportLab, and synchronisation of relevant results with an institutional placement dashboard through a REST API. These components establish a unified workflow for resume processing, immersive interview simulation, preliminary performance feedback, and report generation. However, the evaluation scores remain engineering heuristics, head orientation serves only as a proxy for head alignment rather than direct eye tracking, and formal user studies have not been conducted. Therefore, the system demonstrates the integration of the proposed technical components, while its effectiveness in improving interview confidence and predicting real-world interview performance requires further evaluation.

8.2 Project Achievements

The major achievement of the project is the integration of virtual reality, conversational AI, resume processing, behavioural monitoring, and performance reporting within a single interview-practice framework. The Unity application provides a virtual interview environment with three sequential interviewer stages, while Convai supports conversational interaction with the virtual characters. The Python-based resume-processing module extracts candidate information from PDF documents and prepares a structured profile for the interview workflow. The system forwards available interview transcripts

between stages to maintain conversational context. It also samples headset pose information at 1 Hz and applies a heuristic 35-degree cone to estimate head alignment with the interviewer. Posture stability and response latency are considered as additional behavioural indicators. The evaluation component produces a preliminary performance summary across ten competencies, and the reporting module generates a PDF report using ReportLab. Integration with the institutional placement dashboard through a REST API supports the transfer of relevant interview results. Together, these features demonstrate the practical integration of VR technology, conversational AI services, local resume processing, evaluation logic, and backend reporting functionality.

8.3 Current Limitations

Although the VR Interview Simulation system integrates several functional components, certain limitations remain. The resume-processing module uses text extraction, regular expressions, and keyword-based heuristics, which may not identify every relevant detail in resumes with different formats or layouts. The interview experience depends on the availability and operation of the conversational AI service. The behavioural monitoring module uses headset orientation as an approximate indicator of head alignment rather than direct eye tracking. Similarly, posture stability and response latency provide limited behavioural information and cannot independently establish a candidate's confidence, emotional state, or communication ability. The competency-based evaluation relies on engineering heuristics and has not been validated through formal user studies. Consequently, the scores should be interpreted as preliminary practice feedback rather than reliable predictions of actual interview performance. The technical coding interview track is not implemented in the current version. Further testing is also required to establish usability, scoring consistency, system reliability, and the effectiveness of the framework under realistic interview conditions.

8.4 Future Enhancements

Future development of the VR Interview Simulation system can focus on improving resume understanding, interview personalisation, behavioural analysis, and evaluation reliability. The resume-processing module could be enhanced to handle a wider range of document layouts and extract candidate information more consistently. The interview workflow could be extended with additional interviewer personas, more varied question sets, and improved

context management to provide a broader range of interview scenarios. A technical interview track could also be introduced to support coding-related questions and role-specific technical assessments.

The behavioural monitoring component could be improved through more comprehensive analysis of available headset pose information and, where suitable hardware is available, integration with dedicated eye-tracking technology. Additional indicators could be investigated to provide more detailed feedback on candidate interaction patterns. Any new behavioural measurements would require appropriate validation before being interpreted as indicators of confidence or interview readiness.

The performance evaluation module could be enhanced through clearly defined competency criteria, transparent scoring rules, and systematic validation against human assessments. The PDF reporting module could provide more detailed feedback, competency-wise recommendations, and comparisons across practice sessions. The placement dashboard could also be extended to support interview history, progress tracking, and organised access to candidate reports, subject to appropriate privacy and access controls.

Further experimental evaluation should include user studies involving participants from different backgrounds and levels of interview experience. Such studies could assess usability, interaction quality, feedback usefulness, and changes in interview confidence through suitable evaluation methods. Reliability testing across different VR configurations and conversational scenarios would also help identify technical limitations. These enhancements would build upon the current system and support the development of a more reliable, personalised, and comprehensive platform for immersive interview practice.

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